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Review

The role of m6A RNA methylation in cancer

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ABSTRACT

N6-methyladenosine (m6A), the most abundant internal modification of RNA in eukaryotic cells, has gained increasing attention in recent years. The m6A modification affects multiple aspects of RNA metabolism, ranging from RNA processing, nuclear export, RNA translation to decay. Emerging evidence suggests that m6A methylation plays a critical role in cancer through various mechanisms. Moreover, m6A methylation has provided more possibilities for the early diagnosis and treatment of cancers. In this review, we focus on m6A-associated mechanisms and functions in several major malignancies and summarize the dual role of m6A methylation as well as its prospects in cancer.

1. Introduction

So far, more than 150 types of posttranscriptional modifications have been identified in RNA among all living organisms [1]. N6-methyladenosine (m6A) is the most abundant RNA modification in eukaryotic cells [2,3]. About 0.1–0.4% of all adenosines in isolated RNA are m6A modified in mammals, accounting for approximately 50% of total methylated ribonucleotides [4,5]. m6A RNA methylation mainly occurs at the consensus motif of RRm6ACH ([*G/A/U*][*G/A*]m6AC[*U/A/C*]) [6–10] and it is enriched in 3′ untranslated regions (3′UTRs), around stop codons and within internal long exons [11–13]. Furthermore, m6A also occurs in precursor mRNAs (pre-RNAs) and long noncoding RNAs (lncRNAs) [14–16].

Although early discovery of m6A was in the 1970s, the precise function and mechanism of the m6A modification were largely unclear until recently [17,18]. Similar to DNA and histone methylation, m6A RNA methylation is also dynamic and reversible in mammalian cells and it has been proposed as another form of epigenetic regulation. Emerging evidence suggests that m6A is involved in various aspects of RNA metabolism including pre-mRNA splicing, 3′-end processing, nuclear export, translation regulation, mRNA decay and noncoding RNA (ncRNA) processing [19–22]. m6A methylation has shown various crucial roles in mammals, including embryonic development, neurogenesis, circadian rhythm, stress responses, sex determination and tumorigenesis [23,24].

While still in the beginning stage, the methods developed in recent years have enabled researchers to investigate the implication of m6A methylation in human diseases, especially in cancer. In this review, we discuss how m6A methylation participates in cancer initiation and progression, and the prospects for using m6A methylation as a new diagnostic biomarker and therapeutic target in cancer.

2. m6A writers, erasers and readers

m6A “writers”, “erasers”, and “readers” are proteins that can add, remove, or recognize m6A-modified sites and alter important biological processes accordingly (Fig. 1).

Abbreviations: 3′UTRs, 3′untranslated regions; ADAM19, A disintegrin and metallopeptidase domain 19; ALKBH5, alkB homolog 5; AML, Acute myeloid leukemia; ASB2, Ankyrin repeat and SOCS box containing 2; ATRA, All-trans-retinoic acid; BCL2, B cell lymphoma 2; BCSC, Breast cancer stem cell; CNV, copy number variation; CRC, Colorectal carcinoma; CSCs, Cancer stem cells; CTCs, Circulating tumor cells; DGCR8, DiGeorge Critical Region 8; EGFR, Epidermal growth factor receptor; EMT, Epithelial–mesenchymal transition; FOXM1, Forkhead box protein M1; FTO, Fat mass and obesity-associated protein; GBM, Glioblastoma; GSCs, Glioblastoma stem-like cells; HBXIP, Hepatitis B virus X-interacting protein; HCC, Hepatocellular carcinoma; HIF, Hypoxia-inducible factor; HNRNP, heterogeneous nuclear ribonucleoprotein; HuR, human antigen R; IGF2BP, insulin-like growth factor 2 mRNA-binding protein; KLF4, Kruppel-like factor 4; lncRNA, long noncoding RNA; m6A, N6-Methyladenosine; MA, Meclofenamic acid; METTL, Methyltransferase-like; pre-RNA, precursor mRNA; PTEN, Phosphatase and tensin homolog; R-2HG, R-2-hydroxyglutarate; RARA, Retinoic acid receptor alpha; RBM15, RNA-binding motif protein 15; SAM, S-Adenosyl methionine; SOCS, Suppressor of cytokine signaling; SRSF, Serine and arginine-rich splicing factor; TCGA, The Cancer Genome Atlas; WTAP, Wilms tumor 1-associated protein; XIST, X-inactive specific transcript; YAP, yes-associated protein; YTH, YT521-B homology; ZNF217, Zinc finger protein 217

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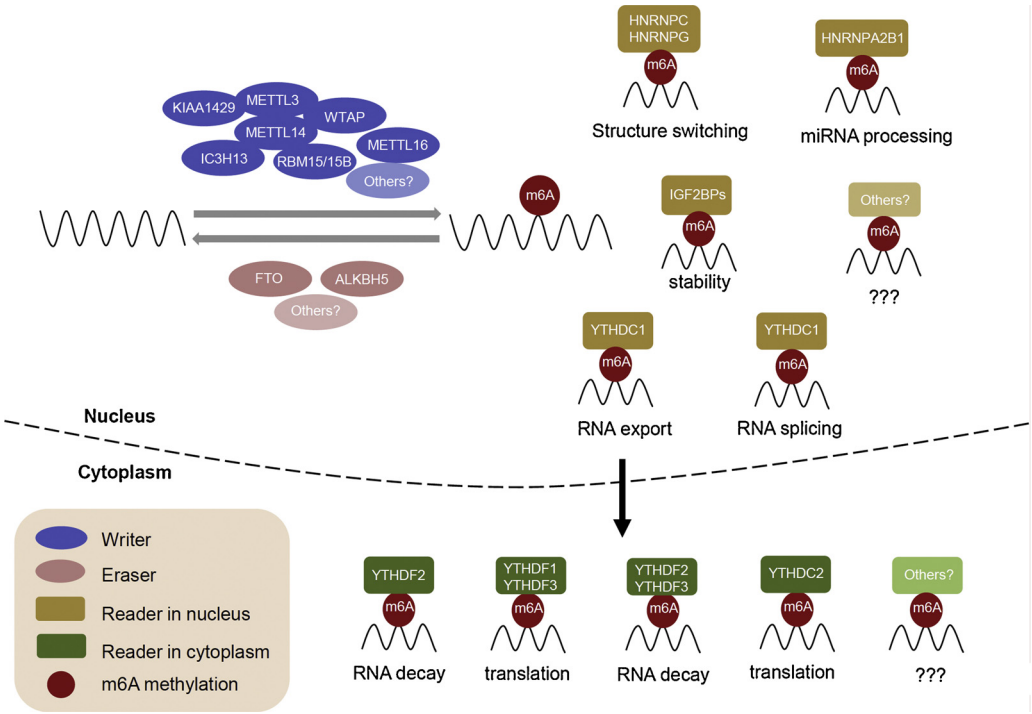

Fig. 1. The process and molecular functions of m6A RNA methylation. m6A RNA methylation is conducted by its writers, erasers, and readers to add, remove, or recognize m6A respectively. Writers refer to the m6A methyltransferases including METTL3, METTL14, WTAP, RBM15/15B, KIAA1429, ZC3H13 and METTL16. Two erasers are demethylases FTO and ALKBH5. Readers are proteins that recognize the m6A sites and performs multiple functions in nucleus or cytoplasm, some of which identified so far are YTHDF1, YTHDF2, YTHDF3, YTHDC1, YTHDC2, and HNRNPC, HNRNPG, HNRNPA2B1 and IGF2BPs. m6A methylation is involved in various aspects of RNA metabolism including RNA splicing, miRNA processing, nuclear export, translation and RNA decay.

2.1. Methyltransferases/Writers

m6A methylation is catalyzed by a multicomponent methyltransferase complex [25]. Methyltransferase-like 3 (METTL3), an S-adenosylmethionine (SAM)-binding protein, is the first characterized component of the m6A methyltransferase complex and is highly conserved in eukaryotes from yeast to human [26]. Methyltransferase-like 14 (METTL14) is another active component of the m6A methyltransferase complex. METTL3 and METTL14 colocalize in nuclear speckles and form a stable heterocomplex in a ratio of 1:1 [27,28]. METTL3 primarily functions as the catalytic core, while METTL14 serves as a structural support for RNA binding [29]. WTAP (Wilms' tumor 1-associated protein) is the third crucial component of the m6A methyltransferase complex. As WTAP lacks a conserved catalytic methylation domain, WTAP itself has no catalytic activity for m6A modification. WTAP works as an adaptor protein that interacts with both METTL3 and METTL14, thus significantly affecting the cellular RNA m6A load [27]. Additionally, WTAP can also interact with many other proteins and lncRNAs [30], suggesting that WTAP may recruit other factors to the methyltransferase complex. Methyltransferase-like 16 (METTL16), the homolog of METTL3, is a newly identified m6A methyltransferase that controls cellular SAM level and methylates U6 small nuclear RNA [31]. Besides, other proteins, such as VIRMA (KIAA1429) [32], RBM15 (RNA-binding motif protein 15) and its paralogue RBM15B [33], and ZC3H13 [34,35], have been found to be the components of the m6A methyltransferase complex and to be required for m6A methylation.

Furthermore, apart from the internal m6A in vertebrate mRNAs, if the transcription start nucleoside is adenosine, its N6 position is methylated to form N6, 2′-O-dimethyladeno-sine (m6Am). A transcriptome-wide analysis revealed that N6-methylation of m6Am promotes the translation of capped mRNAs. A recent study identified PCIF1, a cap-specific adenosine methyltransferase (CAPAM), is responsible for N6-methylation of m6Am [36]. Thus, the methyltransferase to m6Am is different from that of m6A.

2.2. Demethylases/Erasers

In 2011, the first m6A demethylase, fat mass and obesity-associated protein (FTO) was identified, indicating that m6A RNA methylation are reversible and dynamic [37]. Since then, more and more attention has been paid to the possible mechanisms and biological functions of m6A. The second RNA demethylase, α-ketoglutarate-dependent dioxygenase homolog 5 (ALKBH5), was soon discovered in 2013 [38]. Both FTO and ALKBH5 belong to the α-ketoglutarate-dependent dioxygenase family and catalyze m6A demethylation in a Fe(ii) and α-ketoglutarate dependent manner. Either deficiency or overexpression of FTO or ALKBH5 alters the m6A level in cells. In addition, recent studies have found another m6A demethylase, ALKBH3 (AlkB homolog 3) [39]. Intriguingly, ALKBH3 prefers to m6A in tRNA rather than in mRNA or rRNA. Since there are other members of the AlkB subfamily with unknown functions [40], the discovery of more m6A demethylases is unsurprising.

2.3. Readers

The functional interaction between m6A "writers" and "erasers" determines the dynamic and reversible regulation of m6A modification. For the m6A group to have a biological function, it needs to be identified by variable "readers" to exercise different downstream effects [41]. Members of the YT521-B homology (YTH) domain family, including YTHDF1, YTHDF2, YTHDF3, YTHDC1 and YTHDC2, all have a conserved m6A-binding domain and preferentially bind to m6A-modified RNA at the RRm6ACH consensus sequence [42]. YTHDF2, the first characterized m6A reader, accelerates the decay of m6A-modified transcripts by recruiting the CCR4-NOT deadenylase complex directly [43]. In contrast, YTHDF1 was initially demonstrated to bind to m6A sites around the stop codon and then cooperated with the translation initiation machinery to improve the translation efficiency of target RNAs in mammals [44]. YTHDF3 promotes the translation of methylated RNAs in synergy with YTHDF1, while also accelerating mRNA decay by interacting with YTHDF2 directly [20,45]. In conclusion, YTHDF3 functions as the fine-tuning for the RNA accessibility of YTHDF1 and YTHDF2.

The nuclear m6A reader YTHDC1 promotes exon inclusion by recruiting or restricting different splicing factors, such as serine and arginine-rich splicing factors (SRSFs) [46]. YTHDC1 also recognizes m6A methylation on XIST (X-inactive specific transcript) and facilitates XIST-mediated transcriptional repression and X chromosome silencing [33]. Moreover, recent studies have shown that YTHDC1 interacts with SRSF3 and NXF1 to promote the nuclear export of m6A-methylated mRNAs [47], thereby expanding the function of YTHDC1-mediated m6A reading to mRNA metabolism regulation. In addition, YTHDC2 can enhance the translation efficiency and decrease the target mRNA abundance by preferentially binding m6A within the consensus motif [48–51].

Except for the YTH domain family, certain members of the heterogeneous nuclear ribonucleoprotein (HNRNP) family also serve as m6A readers. HNRNPA2B1 was firstly identified as a possible specific m6A-binding protein, promoting primary miRNA processing and maturation by binding m6A-methylated transcripts [52]. Additionally, HNRNPC [53] and HNRNPG [54] also regulate mRNA abundance and splicing by processing m6A-modified RNA transcripts. m6A affects the secondary structure of RNA, which facilitates the binding of transcripts to HNRNPC and HNRNPG and thereby modulating mRNA abundance and splicing [53,54]. This phenomenon is termed “m6A-switch”. Recently, another study confirmed that insulin-like growth factor 2 mRNA-binding proteins (IGF2BPs, including IGF2BP1/2/3) could also recognize m6A modifications, serving as a distinct family of m6A readers [55]. In contrast to the mRNA-decay-promoting function of YTHDF2, IGF2BPs promote the stability and storage of their target mRNAs in an m6A-dependent manner, thus affecting gene regulation and cancer biology.

3. Methods of detecting m6A RNA methylation

The total amount of m6A in RNA can be probed by several methods, including two-dimensional thin layer chromatography [56,57], m6A dot-blot [37] and HPLC-MS/MS (high-performance liquid chromatography-tandem mass spectrometry) [37,38]. However, these approaches are not suitable for widespread identification and localization of modified sites [58].

Before 2012, the transcriptome-wide distribution of m6A was unknown, until MeRIP-seq (methylated RNA immunoprecipitation followed by high-throughput sequencing), also known as m6A-seq (m6A RNA immunoprecipitation sequencing), was developed [11,12]. This method has received considerable attention recently due to its accuracy and reproducibility. In this method, mRNA was fragmented into 100-nt-long oligonucleotides and immunoprecipitated with a specific antibody against m6A. Then immunoprecipitated RNAs were subjected to high-throughput sequencing. As MeRIP-seq relies on RNA fragmentation, the resolution is around 100–200 nt and it is incapable of single-nucleotide resolution detection of methylation sites [11,59].

Other methods have been developed for higher resolution, such as PA-m6A-Seq (photo-crosslinking-assisted m6A-sequencing) and SCARLET (site-specific cleavage and radioactive-labelling followed by ligation-assisted extraction and thin-layer chromatography) [60,61]. However, there are problems to be considered, including time consumption and unsuitability for high throughput applications. Currently, a novel method called miCLIP (m6A individual nucleotide resolution crosslinking immunoprecipitation) has marked a major step forward in the field, which could detect m6A at precise position [8]. In this method, UV induces cross-linking of m6A-containing RNA fragments to m6A antibody, and the antibody-RNA complexes are recovered by affinity purification. Then, reverse transcription of RNA fragments generate mutations or truncations in the cDNA, which are crucial for mapping the precise sites of m6A residues [62].

Furthermore, with the development of CRISPR-based genome engineering, it is currently possible to directly detect the effect of altering any m6A modification site in many organisms. As a supplementary approach, it would be valuable for discovering the function of m6A methylation [22]. Although various methods have been utilized to detect m6A methylation currently, with so many challenges and difficulties, more effective methods are still urgently needed.

4. The role of m6A modification in various cancers

The breakthrough in the discovery and understanding of m6A writers, erasers and readers, together with the development of high-throughput assays, has helped to elucidate the biological functions and the underlying mechanisms of m6A. m6A methylation affects almost every aspect of RNA metabolism, including RNA expression, splicing, nuclear export, translation, decay and RNA-protein interactions [17,18,63,64]. Accumulated studies have revealed that m6A modification in RNA plays a crucial role in many physiological processes, including circadian rhythms, spermatogenesis, embryogenesis, heat shock responses, DNA damage response, pluripotency and reprogramming [65–73]. Furthermore, emerging evidence has indicated that m6A RNA methylation plays an important role in tumor initiation and progression. Moreover, aberrant m6A modification is closely associated with cancer [23,38,74–79]. Herein, we briefly review the recent studies of m6A methylation in various cancers (Table 1, Fig. 2).

4.1. Glioblastoma (GBM)

GBM is the most deadly primary brain tumor. Glioblastoma stem cells (GSCs) are a class of cells with the ability to promote tumor growth and invasion, showing strong resistance to radiotherapy and chemotherapy. The existence of GSCs predicts the poor prognosis of GBM [99]. Studies have shown that m6A RNA methylation is associated with self-renewal of GSCs and tumorigenesis [80]. Downregulation of METTL3 or METTL14 promotes the proliferation and self-renewal of GSCs, while upregulation of METTL3 or inhibition of FTO has the reverse effects. m6A sequencing revealed that downregulation of METTL3 or METTL14 decreases m6A levels on ADAM19 and enhances ADAM19 expression in GSCs, ultimately leading to GBM. MA2, an inhibitor of FTO, has been shown to elevate the m6A level and effectively suppresses GBM progression [80].

Additionally, ALKBH5 has been found to be overexpressed in GSCs and is associated with a poorer prognosis [16]. ALKBH5 knockdown impairs the proliferation of GSCs, which can be rescued by wild-type ALKBH5. Knockdown of ALKBH5 leads to increased m6A levels of FOXM1 nascent transcripts and decreased binding of FOXM1 pre-mRNA with human antigen R (HuR). Recruiting HuR to m6A-modified RNA is essential for the stabilization of FOXM1 mRNA. Furthermore, the lncRNA FOXM1-AS facilitates the effect of ALKBH5 on FOXM1 nascent transcripts and promotes the recruitment of HuR. Thus, ALKBH5 promotes GSC proliferation and GBM progression by enhancing FOXM1 expression through m6A modification [16].

Recently, another study also revealed a crucial role of METTL3-mediated m6A modification in GSC maintenance and radioresistance [85]. This study found that the expression of METTL3 is upregulated in GSCs and attenuated during differentiation. SOX2 was identified as the m6A target of METTL3, and METTL3 promotes the recruitment of HuR to m6A-modified SOX2 mRNA and enhances its stability. GSCs with METTL3 knockdown show enhanced radiosensitivity and reduced DNA repair. METTL3 functions as an oncogene and predicts poor survival in GBM. Thus, abnormal m6A modification in GSCs plays an important role in the development of GBM and is a potential target for GBM treatment.

4.2. Acute myeloid leukemia (AML)

AML is one of the most common hematologic malignancies, with distinct genetic aberrations and unsatisfactory therapeutic effects [100]. It has been reported that METTL3 exerts an oncogenic role in

Table 1
m6A RNA methylation in various cancers.

Molecule	Cancer	Role in cancer	Mechanism	References
METTL3	GBM	Suppressor	Suppresses GSCs growth and self-renewal by enhancing m6A modification of ADAM19 and decreasing its expression	[80]
	AML	Oncogene	Promotes tumorigenesis by enhancing m6A modification and translation of BCL2 and PTEN	[81]
	BC	Oncogene	Promotes tumor growth by enhancing m6A modification and translation of HBXIP	[82]
	HCC	Oncogene	Promotes tumor growth by enhancing m6A modification of SOCS2 and its degradation	[83]
	AML	Oncogene	Binds to the promoter region of mRNA transcript and enhances m6A modification and gene expression	[84]
	GBM	Oncogene	Promotes GSCs growth and self-renewal by upregulating POU3F2, SOX2, SALL2 and OLIG2	[85]
	Endometrial cancer	Suppressor	METTL3 downregulation leads to increased cell proliferation and tumorigenicity through activation of AKT	[86]
METTL14	AML	Oncogene	Promotes tumorigenesis by enhancing m6A modification and expression of MYC and MYB	[87]
	HCC	Suppressor	Suppresses tumor metastasis by promoting miR126 processing in a m6A-dependent manner	[88]
	GBM	Suppressor	Suppresses GSCs growth and self-renewal by enhancing m6A modification of ADAM19 and decreasing its expression	[80]
	Endometrial cancer	Suppressor	METTL14 mutation leads to increased cell proliferation and tumorigenicity through activation of AKT	[86]
FTO	AML	Oncogene	Promotes tumorigenesis by decreasing m6A on ASB2 and RARA at UTRs, which destabilizes ASB2 and RARA	[89]
	AML	Oncogene	Promotes AML carcinogenesis by enhancing MYC and CEBPA mRNA stability	[90]
	GBM	Oncogene	FTO catalytic activity inhibition by MA2 suppresses glioblastoma progression	[80]
	Cervical cancer	Oncogene	Enhances chemoradiotherapy resistance by decreasing m6A modification of β-catenin	[91]
ALKBH5	GBM	Oncogene	Promotes GSCs proliferation and tumor progression by positively regulating FOXM1 through m6A modification	[16]
	BC	Oncogene	ALKBH5 mediates enrichment of BCSCs by decreasing NANOG mRNA methylation and increasing its mRNA levels	[92]
	BC	Oncogene	Promotes tumorigenesis by decreasing m6A methylation of KLF4 and NANOG and enhancing their mRNA stability	[93]
	AML	Suppressor	ALKBH5 copy-number loss is frequent in AML, which predicts poorer survival of AML patients	[94]
	Pancreatic cancer	Suppressor	Suppresses pancreatic cancer motility by decreasing lncRNA KCNK15-AS1 methylation	[95]
YTHDF2	Pancreatic cancer	Dual effect	Promotes tumor proliferation by activating AKT/GSK3β/cyclin D1 and suppresses metastasis by destabilizing YAP mRNA	[96]
	Prostate cancer	Oncogene	Promotes prostate cancer growth and migration by enhancing m6A-containing mRNA degradation	[97]
	HCC	Oncogene	YTHDF2 was upregulated, and miR145 modulates m6A levels by targeting the 3'-UTR of YTHDF2 mRNA in HCC cells	[98]

AML through its methylation catalytic activity [81,84]. Further studies have shown that the expression of METTL3 is more abundant in AML cells than in normal hematopoietic cells. Elevated m6A mediated by METTL3 plays an important role in maintaining pluripotency and inhibiting cell differentiation in AML. METTL3 is recruited by CEBPZ to the promoter region of SP1, which enhances m6A modification of SP1 and stimulates its translation. SP1 then activates the oncogene c-MYC, leading to the development of AML [84]. Moreover, another study

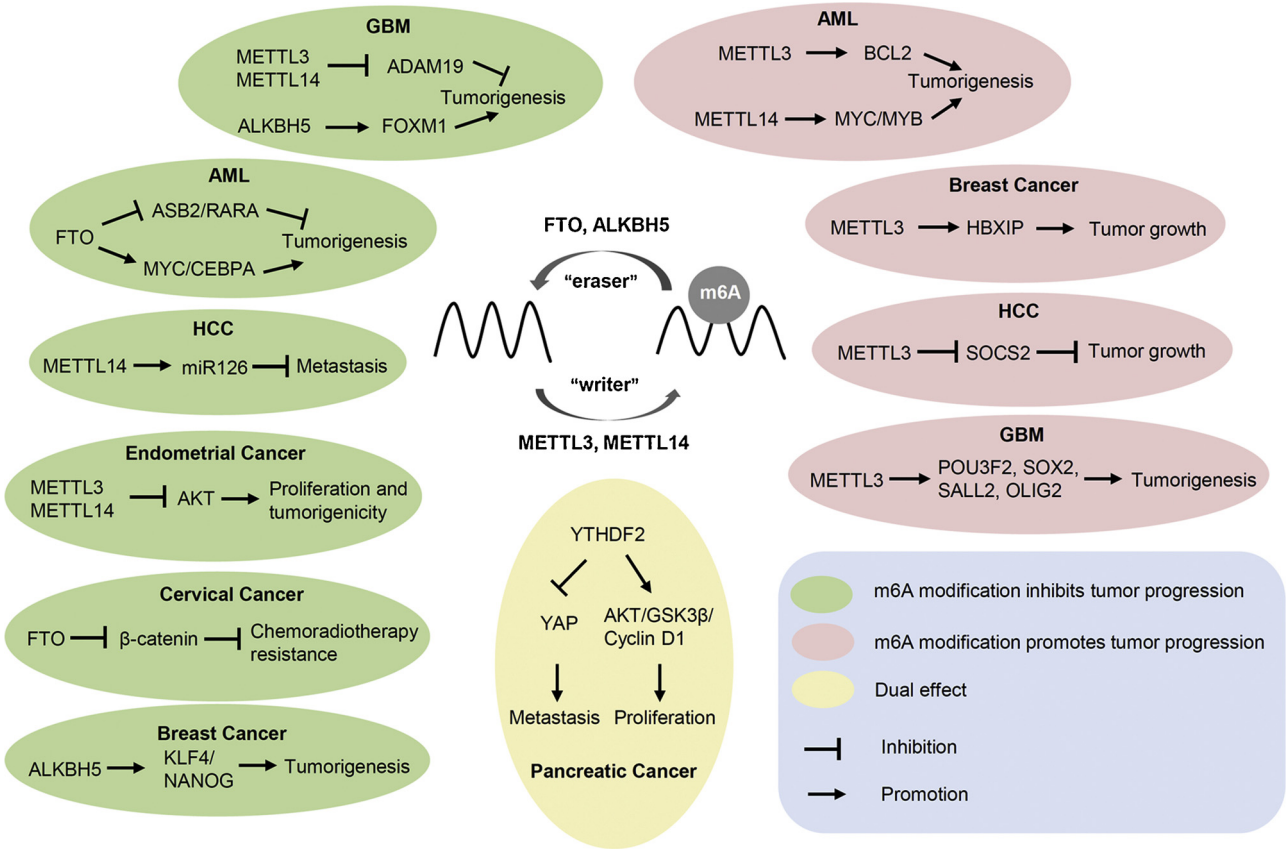

Fig. 2. Multiple functions of m6A methylation in human cancers. Aberrant m6A modification on diverse substrates promotes or suppresses tumour progression. See main text for more details.

found that the upregulated METTL3 levels in AML augments m6A modification of BCL2 and PTEN and facilitates the translation of these mRNAs, which ultimately leads to AML development [81].

Consistent with METTL3, METTL14 is also upregulated in AML cells and was necessary for self-renewal and maintenance of leukemia stem cells [87]. Mechanistically, METTL14 promotes the mRNA stability and translation of the oncogenes MYB and MYC via m6A methylation, while METTL14 itself was negatively regulated by SPI1. Therefore, the SPI1-METTL14-MYB/MYC signaling axis plays a crucial role in AML development [87].

In addition, FTO has been found to exert oncogenic role in hematopoietic cell transformation and AML [89,90]. Li et al. illustrated that FTO was abnormally elevated in AML with MLL-rearrangements, PML-RARA, FLT3-ITD, or NPM1 mutations [89]. Further analysis has confirmed that FTO exerts its oncogenic role by targeting the tumor suppressor ankyrin repeat and SOCS box protein 2 (ASB2) and the retinoic acid receptor alpha (RARA) with its m6A demethylase catalytic activity. Decreased m6A levels in these mRNA transcripts lead to inhibition of ASB2 and RARA expression. ASB2 and RARA are key regulators of normal hematopoiesis and all-trans-retinoic acid (ATRA)-induced differentiation of leukemia cells. Thus, the FTO-mediated inhibition of the ASB2/RARA axis remarkably contributes to leukemogenesis and the response of AML cells to ATRA treatment [89]. Additionally, a recent study demonstrated that R-2-hydroxyglutarate (R-2HG) exhibits antileukemic effects via suppressing FTO [90]. R-2HG could elevate the m6A level of MYC/CEBPA mRNA by inhibiting FTO, which stabilizes MYC/CEBPA mRNA by blocking YTHDF2-mediated RNA decay. Thus, the R-2HG-FTO-m6A axis suppresses the proliferation of leukemia cells by regulating MYC/CEBPA gene expression [90].

FTO is not the only demethylase associated with AML. Mutations and copy number variations (CNVs) of m6A-related genes have been investigated in leukemia patients by analyzing datasets from The Cancer Genome Atlas (TCGA) [94]. This analysis revealed that approximately 2.6% and 10.5% of AML patients carry genetic mutations or CNVs respectively, among which copy number loss of ALKBH5 is the most common. Copy number loss of ALKBH5 is significantly correlated with poor prognosis and p53 mutations in AML [94]. This observation indicates that genetic alterations of m6A regulatory genes may cooperate with p53 mutations in the pathogenesis and/or maintenance of AML.

4.3. Hepatocellular carcinoma (HCC)

HCC is the main type of primary liver cancer, with the fifth highest incidence and the third highest mortality rate of malignant tumors globally [101]. Studies have shown that the progression of HCC is associated with abnormal m6A modifications [83,88,98]. METTL3 is frequently upregulated in HCC patients and contributes to HCC progression [83]. The high expression of METTL3 leads to increased m6A modification of the tumor suppressor SOCS2. YTHDF2 directly recognizes m6A-modified SOCS2 mRNA and subsequently induces the degradation of SOCS2 [83]. Therefore, upregulated METTL3 inhibits the expression of SOCS2 and promotes the development of HCC through m6A methylation.

However, the expression of METTL14 and m6A modification exhibit a converse tendency in the development of HCC. In HCC, particularly in metastatic HCC, decreased m6A level can be observed in HCC tissues, and METTL14 has been indicated to be the main cause of abnormal m6A modification [88]. Downregulation of METTL14 acts as an adverse prognostic factor in HCC patients, and the expression of METTL14 is further decreased in metastatic HCC [88]. METTL14 deficiency enhances the metastatic ability of HCC cells, while METTL14 overexpression inhibits tumor metastasis. Further study has demonstrated that pri-miR126 is a direct substrate of METTL14, and m6A-modified pri-miR126 binds to DGCR8 to positively modulate the maturation of miR126, which has been identified as a tumor suppressor. Thus, METTL14 suppresses HCC metastasis by increasing miR126 levels in a m6A-dependent manner. Additionally, YTHDF2 is upregulated in HCC, and miR145 is a negative regulator of YTHDF2. miRNA145 elevates m6A levels by modulating YTHDF2, which recognizes m6A sites to mediate mRNA degradation [98]. These studies strongly indicate that abnormal m6A methylation is closely related to the development of HCC.

4.4. Breast cancer

Breast cancer has the highest incidence and mortality among all malignant tumors in women [102,103]. Hypoxia, a common feature of the tumor microenvironment, is essential for cancer progression [104,105]. Zhang et al. demonstrated that hypoxia can induce ALKBH5 expression in breast cancer cells in a HIF-dependent manner. ALKBH5 reduces the m6A modification of NANOG mRNA and enhances its stability [92]. NANOG, a pluripotency factor, plays a critical role in the maintenance of cancer stem cells and is necessary for tumor initiation and metastasis. ALKBH5 depletion impairs hypoxia-induced breast cancer stem cell (BCSC) enrichment and tumorigenesis [92]. Consequently, HIF-dependent ALKBH5 expression decreases m6A methylation levels and promotes the enrichment of BCSCs in the hypoxic microenvironment. Further study has shown that hypoxia also induced zinc finger protein 217 (ZNF217) expression in a HIF-dependent manner in breast cancer cells [93]. ZNF217 serves as a m6A methyltransferase inhibitor by sequestering METTL3 and cooperating with ALKBH5 in negatively regulating m6A methylation, ultimately leading to upregulation of NANOG and KLF4 and promoting tumorigenesis [93]. Thus, HIF stimulates pluripotency factor expression and BCSC specification by negatively regulating RNA methylation.

Recently, another study indicated that there is a strongly positive association between the oncoprotein hepatitis B virus X-interacting protein (HBXIP) and METTL3 in the development of breast cancer [82]. Further investigation showed that HBXIP upregulates METTL3 by inhibiting miRNA let-7 g, which downregulates METTL3 expression by targeting its 3′UTR. METTL3 also promotes HBXIP expression through m6A modification. Thus, a positive feedback of HBXIP/let-7 g/METTL3/HBXIP forms, leading to accelerated development of breast cancer [82].

4.5. Pancreatic cancer and prostate cancer

Pancreatic cancer is one of the most aggressive cancers, with a median survival of 5–8 months. A recent study demonstrated that the lncRNA KCNK15-AS1 is significantly downregulated in pancreatic cancer, revealing its inhibitory functions on cell migration and invasion. In addition, ALKBH5 is downregulated in pancreatic cancer, where it can demethylate KCNK15-AS1 and regulate KCNK15-AS1-mediated cell motility [95]. These results reveal a novel mechanism by which ALKBH5 inhibits pancreatic cancer motility by decreasing lncRNA KCNK15-AS1 methylation, identifying a potential therapeutic target for pancreatic cancer. Additionally, another study demonstrated that YTHDF2 is upregulated in pancreatic cancer at both mRNA and protein levels and orchestrates two cellular processes: promoting proliferation and inhibiting epithelial-mesenchymal transition (EMT), a phenomenon called the “migration-proliferation dichotomy” [96]. YTHDF2 promotes tumor proliferation by activating the AKT/GSK3β/cyclin D1 pathway and suppresses migration and invasion by binding to m6A sites of yes-associated protein (YAP) mRNA and destabilizing it [96]. Accordingly, these findings suggest that YTHDF2 may be a new predictive biomarker for pancreatic cancer, but serious consideration is needed to target YTHDF2 for pancreatic cancer therapy.

Additionally, YTHDF2 has also been found to be significantly upregulated in prostate cancer [97]. Knockdown of YTHDF2 greatly elevates m6A levels, and inhibits the proliferation and migration of prostate cancer cells. Further experiments then indicated that miR-493-3p

and YTHDF2 are negatively correlated with each other, and miR-493-3p directly targets the 3′UTR of YTHDF2. Forced expression of miR-493-3p consistently elevates m6A levels and inhibits proliferation and migration [97]. These observations indicate that YTHDF2 and miR-493-3p are involved in prostate cancer development by regulating m6A levels.

4.6. Cervical cancer and endometrial cancer

Cervical cancer is one of the most common gynecologic malignancies, over 90% of which are cervical squamous cell carcinoma (CSCC) [106]. The expression of FTO is significantly elevated in CSCC tissues when compared with that in the respective adjacent normal tissues. FTO enhances chemoradiotherapy resistance through decreasing β-catenin expression by reducing m6A in its mRNA transcripts [91]. Therefore, these findings uncover a critical role of FTO and its substrate m6A modification in the regulation of chemoradiotherapy resistance, suggesting that inhibition of FTO may increase the chemoradiotherapy sensitivity of CSCC.

More recently, Liu et al. found that approximately 70% of endometrial tumors exhibit reductions in m6A methylation when compared with matched normal endometrium. These reductions in m6A methylation are probably due to either METTL14 R298 P mutation or reduced expression of METTL3 [86]. Further study revealed that reduced m6A levels lead to decreased expression of PHLPP2 (a negative AKT regulator) and increased expression of mTORC2 (a positive AKT regulator). Thus, these changes enhance the proliferation and tumorigenicity of endometrial cancer cells through activation of the AKT signaling pathway. Together, these findings reveal that reduced m6A methylation is most likely a carcinogenic mechanism underlying endometrial cancer and suggest that regulation of AKT activity through m6A methylation could be a new direction to explore in the future.

5. The dual role of m6A modification in cancer

5.1. m6A modification inhibits tumor progression

Emerging evidence has demonstrated that downregulation of m6A can promote tumor progression in several types of cancers. In GBM, decreased m6A levels on ADAM19 enhances its expression, promoting GSCs growth and self-renewal and ultimately leading to tumorigenesis [80]. In approximately 70% of endometrial tumors, reduced m6A levels lead to AKT activation and enhanced proliferation and tumorigenicity in endometrial cancer cells [86]. In HCC, the decreased m6A level on pri-miR126 influences its maturation and disrupts the tumor suppressor function of miR126, thereby accelerating tumor progression [88]. The FTO-mediated downregulation of m6A methylation on ASB2 and RARA results in decreased expression of these tumor suppressors, leading to leukemogenesis [89]. Additionally, FTO-mediated downregulation of m6A modification of β-catenin enhances chemoradiotherapy resistance of cervical squamous cell carcinoma [91]. R-2HG exhibits antileukemic effects, since it enhances m6A modification of MYC/CEBPA mRNA by inhibiting FTO [90]. Similarly, inhibition of FTO by MA2 suppresses GBM progression [80]. In GSCs, reduced m6A modification of FOXM1 mediated by ALKBH5 promotes GSCs proliferation and tumor progression [16]. Additionally, decreased m6A modification stabilize KLF4 and NANOG RNA and ultimately contributes to the enrichment of BCSCs and tumorigenesis in breast cancer [92,93]. In summary, decreased RNA methylation may participate in tumor development and progression. Therefore, the regulation of the transcription of these genes via enhancing m6A modification may be a novel therapeutic strategy for cancer.

5.2. m6A modification promotes tumor progression

In AML, elevated m6A levels on SP1, MYB, MYC, BCL2 and PTEN promote the stability and translation of these mRNAs, leading to tumorigenesis and the development of AML [81,84,87]. In HCC, the enhanced m6A modification of SOCS2 promotes its degradation, which accelerates tumor progression [83]. Additionally, in breast cancer, upregulated m6A methylation of HBXIP accelerates tumorigenesis [82]. Through TCGA datasets analysis, ALKBH5 copy number loss has been found to be frequent and notably associated with p53 mutations in AML, which predicts poorer prognosis of AML patients [94]. In conclusion, RNA methylation leads to certain alterations in tumor-specific oncogene expression and biological function, thereby promoting tumor progression.

6. Conclusions and future directions

The rapid development of m6A RNA methylation research contributes to revealing underlying mechanisms of cancer initiation and progression. As the most prevalent modification of RNA, m6A methylation is a double-edged sword for cancer: over m6A modification of certain genes could lead to alterations of mRNA behavior and expression, resulting in the acceleration of tumor development, whereas lacking of m6A modification on other genes may also lead to tumor progression.

The critical role of m6A RNA methylation in cancer initiation and progression provides new possibilities for the early diagnosis and treatment of cancers. Huang et al. have found a remarkable upregulation in m6A modification in circulating tumor cells (CTCs) from lung cancer patients, suggesting that upregulated m6A RNA methylation in CTCs may contribute to monitoring and preventing tumor metastasis [110]. Moreover, it has been reported that the regulators or inhibitors of the m6A may have therapeutic potential in cancer. MA2, the ethyl ester form of meclofenamic acid (MA), has been shown to effectively suppresses GBM progression by inhibiting FTO [80]. Additionally, R-2HG exhibits antileukemic effects, since it increases m6A modification in sensitive cells via suppressing FTO [90].

Furthermore, the modification of m6A regulators in cancers has been reported. Du et al. found that METTL3 is modified by SUMO1 and SUMOylation of METTL3 significantly represses its m6A methyltransferase activity resulting in the decrease of m6A levels in mRNAs [107]. Moreover, SUMOylation of METTL3 also can promote tumor growth in human non-small cell lung carcinoma cells. These results suggest that modification of m6A regulators maybe a novel molecular mechanism underlying regulation of m6A methylation.

However, there remains numerous challenges in m6A RNA methylation research. Firstly, it is worth noting that m6A-related proteins may affect tumor progression in a m6A-independent manner. For instance, in lung cancer, upregulated METTL3 promotes growth and invasion of cancer cells by promoting the translation of target transcripts (e.g., EGFR and TAZ) independent of its catalytic activity [108]. In addition, WTAP has been described as a novel oncogenic protein in AML, but further studies are needed to determine whether this function is related to its role as a cofactor of the m6A methyltransferase complex [109].

Secondly, little is known about the mechanisms that control the expressions and activities of the different m6A writers, erasers and readers in the different tissues, resulting in their cancer promoting or suppressing effects. For example, METTL3 has different expression and activity in different tumor tissues. Cai et al. found that HBXIP upregulated METTL3 by inhibiting miRNA let-7 g in breast cancer cells, leading to accelerated development of breast cancer. However, how METTL3 are regulated in other cancers remains unclear.

Thirdly, further studies are needed to assess the clinical value of m6A in human cancers. Given the dual role of m6A in cancer promotion and suppression, such as METTL3, which acts as an oncogene in AML but as a tumor suppressor in GBM, inhibiting a specific m6A regulator may induce the formation of another type of cancer or other diseases. Even if a certain m6A regulator, such as FTO, plays an oncogene role in

a variety of cancers according to current studies, we cannot assume that it plays the same role in all cancers. Therefore, further studies are needed before applying FTO inhibitors to cancer therapy.

Conflicts of interest

No potential conflicts of interest are disclosed.

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